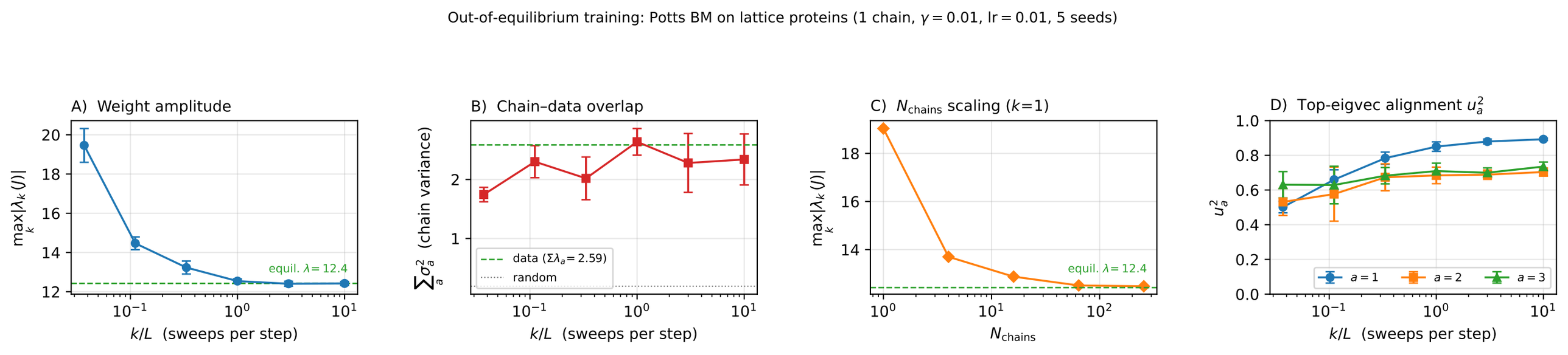
Out-of-equilibrium training: Potts BM on lattice proteins (1 chain, $\gamma = 0.01$, lr $= 0.01$, 5 seeds)
A) Weight amplitude
$\max_k |\lambda_k(J)|$
equil. $\lambda = 12.4$
k/L (sweeps per step)
B) Chain–data overlap
$\sum_a \sigma_a^2$ (chain variance)
data ($\Sigma\lambda_a = 2.59$)
random
k/L (sweeps per step)
C) N_{chains} scaling (k=1)
$\max_k |\lambda_k(J)|$
equil. $\lambda = 12.4$
N_{chains}
D) Top-eigvec alignment u_a^2
u_a^2
$a = 1$
$a = 2$
$a = 3$
k/L (sweeps per step)